

Effects of variable viscosity and thermal diffusivity on MHD free convection flow along a moving vertical plate embedded in a porous medium with heat generation

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Abstract - The present paper is an effort to deal with the problem of the influence of variable viscosity and thermal diffusivity on the MHD free convection flow along a moving vertical porous plate embedded in a porous medium with heat generation. The governing partial differential equations of the flow field are transformed into ordinary differential equations by means of similarity transformation. The resultant equations are solved numerically using Runge-Kutta fourth order method along with shooting technique. The effects of variable thermo-viscous parameters, magnetic parameter, permeability parameter and suction parameter on the velocity, temperature, skin-friction coefficient and Nusselt number are obtained and discussed in detail. The skin-friction coefficient decreases as the magnetic parameter or suction parameter increases. The Nusselt number decreases as β or magnetic parameter or permeability parameter or Grashof number.

Key Words- variable viscosity, thermal diffusivity, MHD, porous medium and heat generation.

1. INTRODUCTION

Natural convection flows driven by temperature differences are of great interest in a number of industrial applications. Buoyancy is also of importance in an environment where differences between land and air temperature can give rise to complicated flow patterns, and in enclosures such as ventilated and heated rooms and reactor configurations. Natural convection flows driven by temperature differences have been studied extensively. For example, Pohlhausen [1] first studied the steady free convection flow past a semi-infinite vertical plate by integral method. But the similarity solution to steady free convection flow past a semi-infinite vertical plate was presented by Ostrach [2], who solved the ordinary non-linear equations by a numerical method.

Siegel [3] was the first to study the transient free convection flow past a semi-infinite vertical plate by integral method. The same problem was studied by Gebhart [4] by an approximate method.

In all the above studies, the free convection flow along a vertical flat plate was restricted, in general, to the case where the temperature difference between the plate and the fluid is small, so that the fluid properties may be taken as constant. For the fluids, which are important in the theory of lubrication, the heat generated by the internal friction and the corresponding rise in temperature do affect the viscosity and thermal diffusivity of the fluid and they can no longer be regarded as constant. The physical properties of fluids such as viscosity and thermal diffusivity may change significantly with temperature. The temperature dependent property problem is further complicated by the fact that the properties of different fluids behave differently with temperature. Different relations between the physical properties of fluids and temperature are given by Kays and Crawford [5]. Mehta and Sood [6] have shown that when this effect is included, the flow characteristics may be substantially changed compared to the constant viscosity case. The influence of variable viscosity on the laminar boundary layer flow and heat transfer due to a continuously moving flat plate is examined by Pop et al. [7]. Kafousias and Williams [8] investigated the effect of temperature-dependent viscosity on free-forced convective laminar boundary layer flow past a vertical isothermal plate. Hossain and Munir [9] analyzed a two-dimensional mixed convection flow of a viscous incompressible fluid of temperature dependent viscosity past a vertical plate.

Elbashbeshy and Ibrahim [10] investigated the steady free convection flow with variable viscosity